

# Task 1: Language Modeling Basics (10 points)

## 1. Language Modeling

### Definition:

Language modeling is the task of assigning a probability to a sequence of tokens (words, subwords, or characters).

Formally, for a sequence  $w_1, w_2, \dots, w_T$ , a language model estimates:

$$P(w_1, w_2, \dots, w_T)$$

This probability reflects how likely a sequence is according to patterns learned from data.

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## 2. Left-to-right framework

Using the **chain rule of probability**, the joint probability of a sequence is factorized as:

$$P(w_1, \dots, w_T) = \prod_{t=1}^T P(w_t | w_1, \dots, w_{t-1})$$

This is called **left-to-right factorization**, because each token is predicted using all previous tokens as context.

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## 3. Next-token prediction

**Next-token prediction** means learning to predict the probability distribution of the next token given the previous tokens:

$$P(w_t | w_1, \dots, w_{t-1})$$

**Why it is the standard objective:**

- It directly follows from the chain rule
  - It allows training on raw text without labels
  - It generalizes to many tasks (generation, completion, scoring)
  - It is computationally efficient and scalable
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## 4. Out-of-Vocabulary (OOV)

**OOV** refers to words that are not present in the model's vocabulary and therefore cannot be directly represented.

**Ways to handle OOV:**

1. **<UNK> token** – replace unseen words with a special unknown token
  2. **Subword tokenization** (BPE, WordPiece, SentencePiece) – split rare words into smaller units
  3. **Character-level models** – model text as sequences of characters (no OOV, but longer sequences)
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## 5. Applications of Language Models

1. **Automatic Speech Recognition (ASR)**  
LM scores candidate transcriptions and helps choose the most fluent sentence.
2. **Machine Translation (MT)**  
LM ensures grammatical and fluent target-language output.
3. **Autocomplete / Text prediction**  
LM predicts the most likely next word while typing.